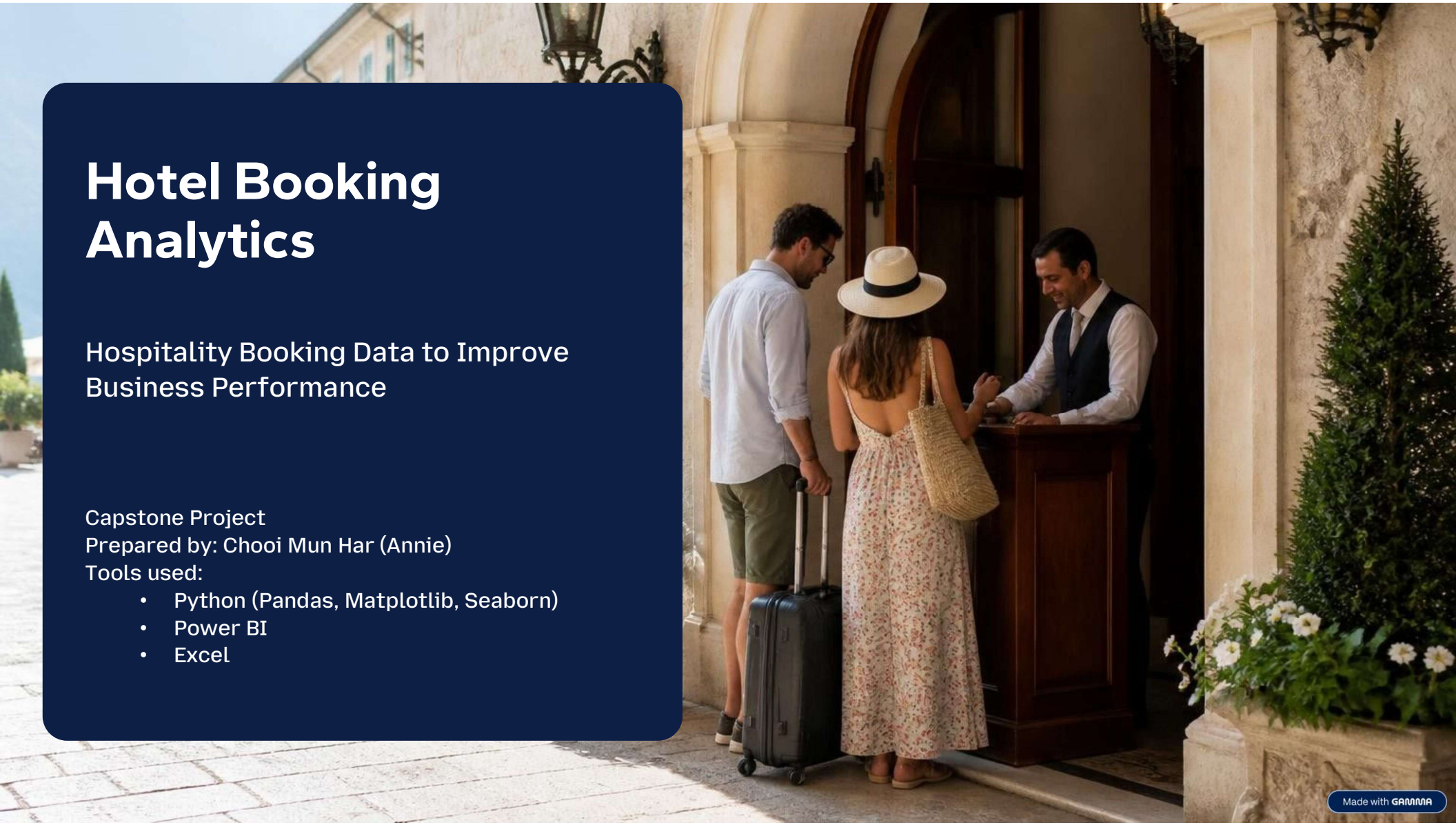


Hotel Booking Analytics

Hospitality Booking Data to Improve Business Performance

Capstone Project
Prepared by: Chooi Mun Har (Annie)
Tools used:

- Python (Pandas, Matplotlib, Seaborn)
- Power BI
- Excel

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Project Background

Business Context

The hospitality industry faces challenges in managing booking cancellations, maximizing occupancy rates, improving customer retention, and optimizing revenue generation.

Business Problem

How can hospitality booking data be leveraged to identify the key factors influencing booking cancellations, occupancy patterns, customer value, distribution channel effectiveness, and reservation outcomes in order to support data-driven decision-making and improve hotel performance?

Project Goal

To transform historical hotel booking data into actionable business insights that support strategic decision-making and improve overall hotel performance.



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Business Problem & Objectives

Business Problem

How can hospitality booking data be leveraged to identify the key factors influencing booking cancellations, occupancy patterns, customer value, distribution channel effectiveness, and reservation outcomes in order to support data-driven decision-making and improve hotel performance?

Research Objectives



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Dataset Summary

Dataset Size

- Data consists of historical arrivals: July 2015–August 2017
- Total Records: 119,390 bookings
- Hotel Types:
  - ✓ City Hotel
  - ✓ Resort Hotel

Key Data Categories

- Customer Information
  - ✓ Customer Type
  - ✓ Country
- Booking Information
  - ✓ Lead Time
  - ✓ Deposit Type
  - ✓ Market Segment
- Revenue Information
  - ✓ ADR (Average Daily Rate)
- Reservation Information
  - ✓ Reservation Status
  - ✓ Special Requests
- Distribution Information
  - ✓ Distribution Channel

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Data Preparation

Data Cleaning and Transformations

Steps performed in Python:

Column Name	Original Type	Updated Type	Action Taken
children	float64	int64	Filled missing values with 0
agent	float64	Int64	Filled missing values with 0 (No Agent)
company	float64	Int64	Filled missing values with 0 (No Company)
country	object	object	Missing values replaced with 'Unknown'
reservation_status_date	object	datetime	Converted to date format
is_canceled	int64	Boolean	Converted 0/1 to Boolean values
is_repeated_guest	int64	Boolean	Converted 0/1 to Boolean values
arrival_date	Derived	datetime	Created from year, month, and day of month fields
total_nights	Derived	int64	Total stay duration created from Weekend Nights + Week Nights
total_price	Derived	float64	Estimated booking value created from ADR × Total Nights

Additional transformations performed in PowerBI :

- Lead Time Categories created for segmentation analysis
- DAX measures created for Average ADR, Average Stay Length, Booking Count, Cancellation Rate, Occupancy Rate, Total Bookings and Total Revenue.

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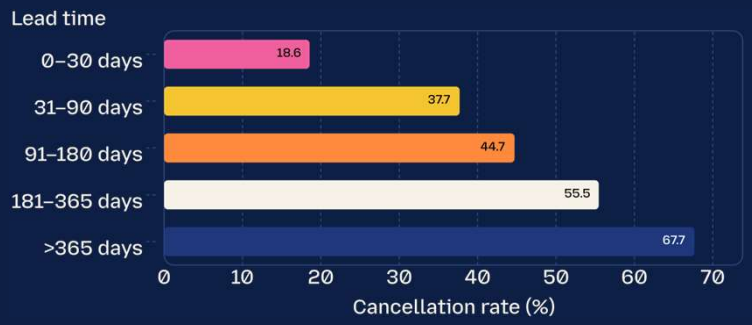
1 - Factors Influencing Booking Cancellations

- Cancellation risk rises with booking lead time

67.66%

cancellation rate for bookings made over 365 days in advance

For 0–30 days, the cancellation rate is 18.56%.



- Deposit and customer mix reveal different kinds of risk

99.36% — **Non-Refund** cancellation rate within the deposit category.

40.75% — **Transient** cancellations highest rate among all customer segments.

10.23% — **Group** cancellations lowest cancellations rate among all customer segments.

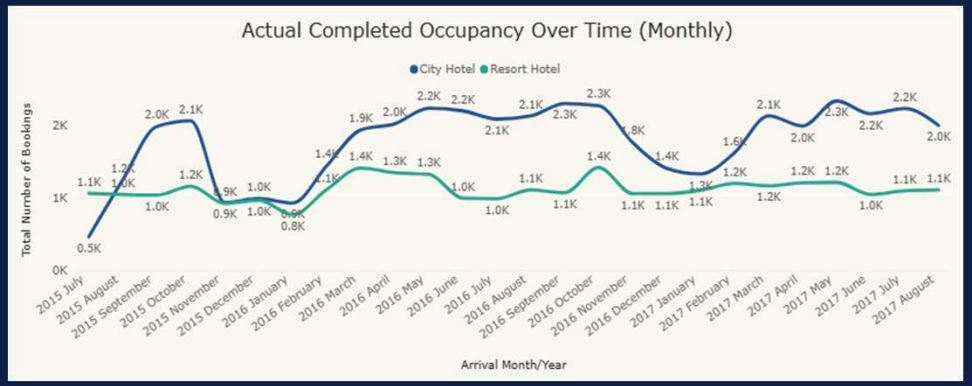
Cancellation Rate by Customer Type & Deposit Type

customer_type	No Deposit	Non Refund	Refundable	Total
Contract	20.28%	100.00%	100.00%	30.96%
Group	10.37%			10.23%
Transient	30.76%	100.00%	100.00%	40.75%
Transient-Party	22.36%	91.80%	10.61%	25.43%
Total	28.38%	99.36%	22.22%	37.04%

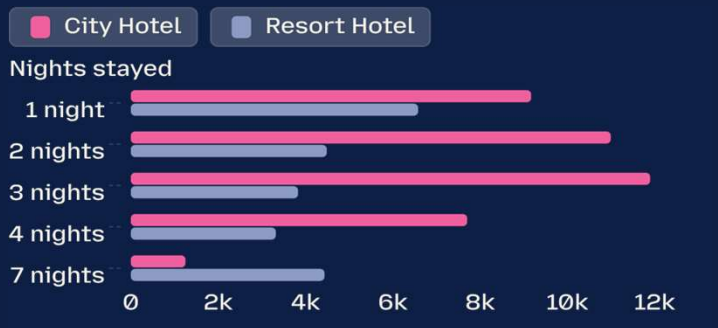
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2 – Occupancy Pattern Analysis

- City Hotel consistently achieved higher occupancy



- Short stays dominate; resort guests stay longer



Completed bookings by stay length

Peak City Hotel

- 2,339 bookings (May 2017)

Peak Resort Hotel

- 1,417 bookings (Oct 2016)

Highest Occupancy Segment

- City Hotel

3.5× more seven-night Resort bookings

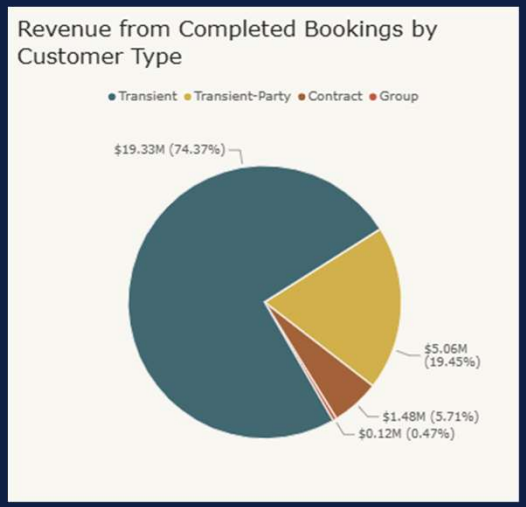
Resort: 4,435; City: 1,251. City peaks at 11,895 bookings for 3 nights.

City bookings cluster in 1–4 nights; Resort has a larger relative share of longer stays.

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3 – Customer Segment Evaluation

- Transient customers generate the highest revenue but carry the greatest cancellation risk





Cancellation rates: Transient 40.75%; Contract 30.96%; Transient-Party 25.43%; Group 10.23%.

4× more revenue

Transient

- Highest revenue ($19.33M)
- Highest ADR ($104.78)
- Highest cancellation rate (40.75%)

Transient-Party

- Second-largest revenue ($5.06M)
- Moderate cancellation rate (25.43%)

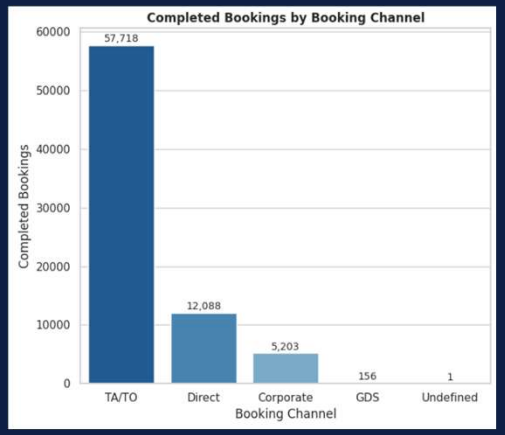
Group

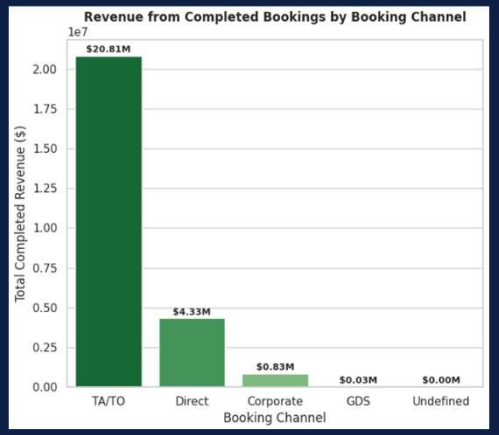
- Lowest revenue ($0.12M)
- Lowest cancellation rate (10.23%)

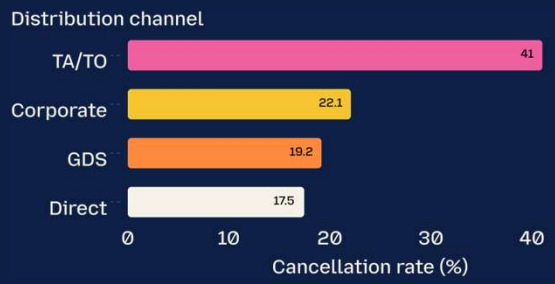
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4 - Distribution Channel Performance

- TA/TO drives scale while Direct delivers the most reliable bookings







TA/TO

- Highest completed bookings (57,718)
- Highest realized revenue ($20.81M)
- Highest cancellation rate (41.03%)

Direct

- Lowest cancellation rate (17.46%)
- Strong revenue ($4.33M)
- Most reliable source of completed stays

Corporate

- Moderate volume and revenue
- More stable than TA/TO

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5 – Reservation Outcome Analysis

- Completed stays show more guest engagement



Average lead time: Canceled 147.3 days; Check-Out 79.98; No-Show 57.18.

Bookings: 43,017 Canceled; 75,166 Check-Out; 1,207 No-Show.

0.71 vs 0.32

Average special requests: Check-Out vs Canceled

Check-Out

- 75,166 bookings, the majority outcome, with the highest average special requests at 0.71.

Cancelled

- Average lead time of 147.3 days, nearly twice completed stays at 79.9 days.

No-Show

- 1,207 bookings, with shorter lead times and moderate request levels.

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Recommended Strategy

- Prioritize actions that improve reliability without suppressing high-value demand.

Priority	Initiative	Target	Action	KPI
1	Risk-tiered booking controls	Long lead time + Transient + OTA/TA/TO	Deposit or guarantee rules, reminder cadence, reconfirmation	Cancellation rate by risk tier
2	Direct conversion program	Repeat and engaged guests	Member-only value, post-stay offer, direct rebooking	Direct share and direct cancellation rate
3	Seasonal occupancy playbook	May and August peaks; hotel-type differences	Staffing, inventory, minimum-stay and package rules	Occupancy, ADR and stay length
4	Engagement-to-arrival workflow	Guests with few or no requests	Pre-arrival preference capture and service prompts	Request rate and stay completion

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Hotel Booking Performance Dashboard

Data from July 2015 to Aug 2017

$26M	75K	62.96%	37.04%
Total Completed Revenue	Total Completed Bookings	Occupancy Rate	Cancellation Rate

Actual Completed Occupancy Over Time (Monthly)



Revenue Contribution by Distribution Channel



Customer Type: All

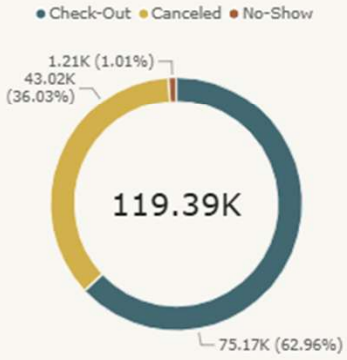
Hotel: All

Market Segment: All

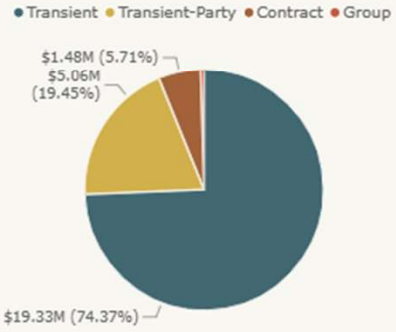
Reservation Status: All

Arrival Date: 1/7/2015 – 31/8/2017

Reservation Outcomes



Revenue from Completed Bookings by Customer Type



Cancellation Rate by Customer Type & Deposit Type

customer_type	No Deposit	Non Refund	Refundable	Total
Contract	20.28%	100.00%	100.00%	30.96%
Group	10.37%			10.23%
Transient	30.76%	100.00%	100.00%	40.75%
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Total	28.38%	99.36%	22.22%	37.04%

Cancellation Rate by Lead Time



The management decision

Grow the demand that stays.

Use risk-tiered controls for high-value volatile bookings, shift repeatable demand toward Direct, and operationalize seasonal occupancy patterns.

Protect
high-revenue demand

Convert
engaged guests to Direct

Plan
by season and hotel type

Measure
completion, not bookings alone

Q&A

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